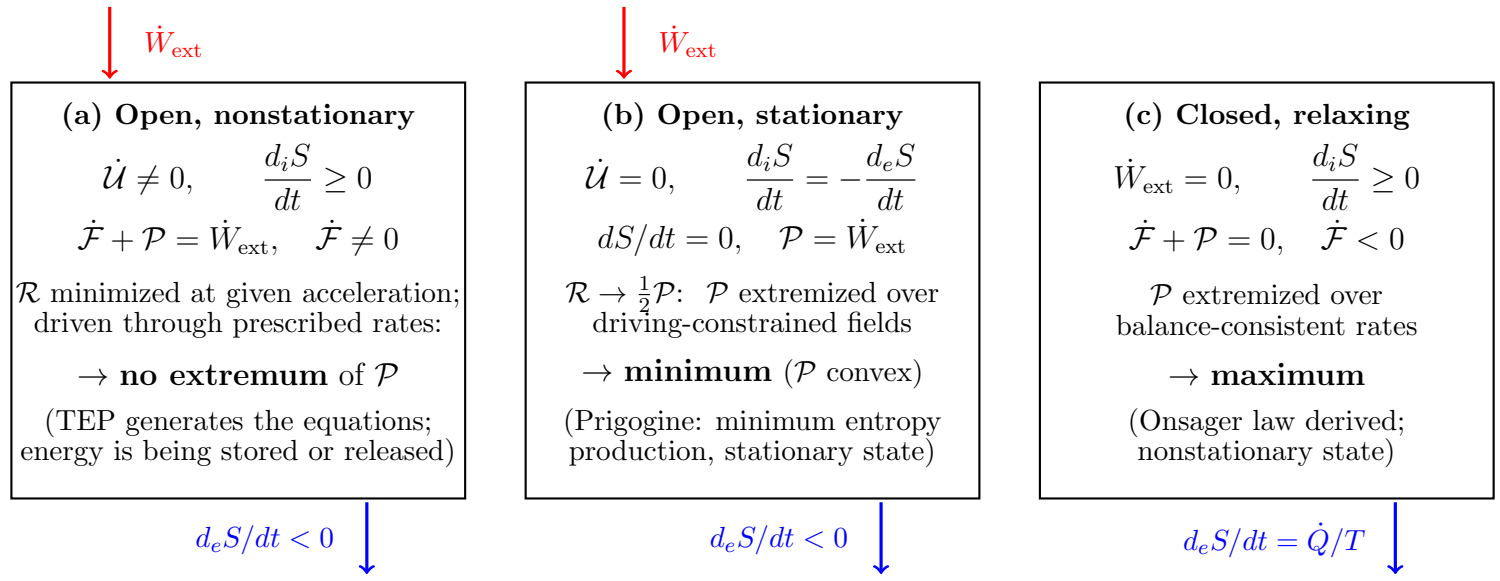

$\dot{W}_{\text{ext}}$
(a) Open, nonstationary
$\dot{\mathcal{U}} \neq 0, \quad \frac{d_i S}{dt} \geq 0$
$\dot{\mathcal{F}} + \mathcal{P} = \dot{W}_{\text{ext}}, \quad \dot{\mathcal{F}} \neq 0$
$\mathcal{R}$ minimized at given acceleration; driven through prescribed rates:
$\rightarrow$ no extremum of $\mathcal{P}$
(TEP generates the equations; energy is being stored or released)
$d_e S/dt < 0$
$\dot{W}_{\text{ext}}$
(b) Open, stationary
$\dot{\mathcal{U}} = 0, \quad \frac{d_i S}{dt} = -\frac{d_e S}{dt}$
$dS/dt = 0, \quad \mathcal{P} = \dot{W}_{\text{ext}}$
$\mathcal{R} \rightarrow \frac{1}{2}\mathcal{P}$: $\mathcal{P}$ extremized over driving-constrained fields
$\rightarrow$ minimum ($\mathcal{P}$ convex)
(Prigogine: minimum entropy production, stationary state)
$d_e S/dt < 0$
(c) Closed, relaxing
$\dot{W}_{\text{ext}} = 0, \quad \frac{d_i S}{dt} \geq 0$
$\dot{\mathcal{F}} + \mathcal{P} = 0, \quad \dot{\mathcal{F}} < 0$
$\mathcal{P}$ extremized over balance-consistent rates
$\rightarrow$ maximum
(Onsager law derived; nonstationary state)
$d_e S/dt = \dot{Q}/T$